Pushkar Patil

AI/ML ENGINEER

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SUMMARY

AI/ML Engineer with 3+ years of experience across healthcare, manufacturing, and research data environments. Build prediction models, NLP systems, and reporting pipelines using cloud platforms like AWS, GCP, and Azure to analyze large datasets, support experiments, and improve clinical, operational, and business decisions. Used Python, SQL, Spark, Power BI, Tableau, and Excel to analyze patient and operational data, support experimentation, and improve reporting and decision-making.

SKILLS

Analytics & Statistics: Statistical Analysis, Hypothesis Testing, A/B Testing, p-values, Confidence Intervals, Statistical Significance, Sampling Methods, Experiment Design, Data Validation, Reporting, KPI Analysis

Machine Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Anomaly Detection, Predictive Modeling, Time Series Forecasting

NLP & LLMs: Transformers, BERT, Hugging Face, Named Entity Recognition, Text Classification, Retrieval Augmented Generation (RAG), Embeddings, Vector Search, Semantic Search

Data Engineering: PySpark, Apache Spark, Spark SQL, Airflow, Flink, ETL Pipelines, Data Streaming, Data Modeling, Data Marts, Monitoring

MLOps & Deployment: MLflow, Docker, Kubernetes, CI/CD, Model Monitoring, Model Versioning

Databases: Snowflake, PostgreSQL, BigQuery, MongoDB, Redis

Visualization: Power BI, Tableau, Looker, Excel, Plotly

Enterprise & Domain Tools: SAP, Fetch, Agile PLM, Material Master Data, BOMs, Routings, MRP, ATP, cGMP Manufacturing Support

EDUCATION

Master of Science in Information Systems **Dec 2024**

Northeastern University, Boston, MA

Bachelor of Science in Computer Science **Oct 2020**

Mumbai University, Mumbai, IN

EXPERIENCE

Abbott Laboratories, IN | AI Engineer **Feb 2025 - Present**

- Engineered a GMP manufacturing risk prediction pipeline using Python, PySpark, PyTorch, and Scikit-learn to analyze 50M+ batch, process, and quality records, improving early detection of production deviations by 21% across manufacturing monitoring systems.
- Designed transformer-based NLP models using BERT Hugging Face to extract deviation, CAPA, batch record entities from unstructured quality manufacturing documents, increasing data availability for downstream compliance and process analytics by 38%.
- Implemented a Retrieval Augmented Generation (RAG) pipeline with embeddings and vector search, enabling semantic retrieval across 10M+ GMP documents including SOPs, batch records, validation reports, and investigation logs, reducing research and review time by 42%.
- Optimized anomaly detection models using XGBoost and LightGBM on streaming equipment and process data, identifying abnormal manufacturing patterns, and improving critical alert detection rates for GMP operations by 27%.
- Analyzed model and process results using SQL and Python, applying hypothesis testing, p-values, and confidence intervals to validate quality trends and support reporting for manufacturing, quality, and validation stakeholders.
- Supported experiment design for healthcare digital tools by defining test groups, reviewing sample quality, and presenting A/B testing results through Excel, Power BI, and Tableau dashboards.

Experiential AI, MA | Data Analyst/Engineer **Jan 2024 - Aug 2024**

- Streamlined 10+ enterprise workflows by embedding AI orchestration layers with n8n and Make.com, incorporating geohash based location search for spatial automations and reclaiming 15 engineer hours/week.
- Engineered a hyperlocal environmental intelligence platform on Flask by ingesting sensor telemetry via GCP Pub/Sub, Flink Clusters, Big Query, sustaining 1M+ records/minute throughput with millisecond-level SLAs.
- Constructed high performance research data marts with geohash partitioning and location-based query acceleration on Big Query and Spark SQL, enabling near real time geo analytics for 10M+ users.
- Delivered real-time & historical insights to government stakeholders using Looker, ArcGIS dashboards powered by federated query engines, directly reducing reporting latency and operational cost by $10K/month.

Asfaliea, India | Data Engineer I **Dec 2020 – Aug 2022**

- Managed material master data for 50K+ raw materials, packaging components, and finished goods across two healthcare manufacturing sites in SAP, improving planning accuracy and supporting compliant production execution.
- Maintained BOMs, production versions, routings, and planning parameters for regulated manufacturing processes, enabling reliable batch scheduling, material staging, and short-term capacity planning under cGMP requirements.
- Reviewed MRP outputs, ATP availability, and shortage signals to resolve material gaps before they affected production orders, purchase requisitions, or patient-critical supply commitments.
- Led 90+ Agile PLM change orders and change actions tied to material, packaging, process updates, strengthening documentation control, traceability, and audit readiness across manufacturing and quality teams
- Built AWS-based reporting pipelines using S3, AWS Glue with Spark, Lambda, EC2, ECS to consolidate SAP, inventory, procurement, and production data, reducing operational reporting latency by 40%.
- Created a daily distressed-inventory tracker with DynamoDB, Lambda, SQL, Power BI to monitor expiring, blocked, and slow-moving materials, helping managers prioritize actions and cut issue resolution time by 35%.
- Partnered with Planning, Procurement, Manufacturing, Quality, Finance, IT to align demand, material readiness, supplier timelines, capacity constraints into executable production plans, while reducing manual reporting effort by 65% through Power BI and Excel reporting.

PROJECTS

LogiFlow: Logistics Data Pipeline | GCP, Pulumi, FastAPI, Apache Spark, Docker, Airflow

- Formulated a logistics analytics pipeline on GCP using Python, Dataflow, and BigQuery, increasing throughput by 40%.
- Automated workflows by integrating Mage.ai for data validation within Cloud Scheduler, Dataflow pipelines, improving analytics accuracy by 35%.

PodcastGPT: RAG-Based AI for Intelligent Podcast Search | FAISS, OpenAI API, LangChain, FastAPI

- Innovated PodcastGPT by integrating FAISS, OpenAI GPT, and LangChain, and incorporated GitHub Actions for CI/CD to accelerate query retrieval speed by 15%.
- Orchestrated a scalable API using FastAPI, Docker, AWS Lambda, and Step Functions to optimize podcast search efficiency.